

Invariant Planes, Exact Motion, and Null Degeneration of a Constant Electromagnetic Lorentz Flow

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Abstract

We classify proper-time Lorentz motion in every constant electromagnetic field. The characteristic polynomial produces complementary boost and rotation planes, an exact exponential, and an exact position integral. We then classify velocity periods, prove physical worldline nonclosure, and resolve electric, magnetic, and secular faces. Proper time is never replaced by coordinate time.

1 Convention and invariant factorization

Use $\eta = \text{diag}(1, -1, -1, -1)$ and $\kappa = q/m$. We freeze

$$A = \kappa \begin{pmatrix} 0 & E_x & E_y & E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (1)$$

Thus $A^T \eta + \eta A = 0$, and for $u = (u^0, u^0 \mathbf{v})$ the spatial part of $u' = Au$ is $\mathbf{u}' = \kappa u^0 (\mathbf{E} + \mathbf{v} \times \mathbf{B})$. This fixes both electric and magnetic signs. The prime denotes proper time τ , and $x' = u$. Exact constant-field Lorentz evolution has a long lineage; see Chin [1]. We make no priority claim.

Theorem 1 (invariant scales and planes). *There are unique $a, b \geq 0$ such that*

$$\chi_A(z) = (z^2 - a^2)(z^2 + b^2), \quad (2)$$

$$a^2 - b^2 = \kappa^2(|\mathbf{E}|^2 - |\mathbf{B}|^2), \quad a^2 b^2 = \kappa^4(\mathbf{E} \cdot \mathbf{B})^2. \quad (3)$$

If $D = a^2 + b^2 > 0$, then

$$P_h = \frac{A^2 + b^2 I}{D}, \quad P_r = \frac{-A^2 + a^2 I}{D} \quad (4)$$

are complementary projectors, with $A^2 P_h = a^2 P_h$ and $A^2 P_r = -b^2 P_r$.

Proof. A direct determinant of (1) gives $z^4 + \kappa^2(|\mathbf{B}|^2 - |\mathbf{E}|^2)z^2 - \kappa^4(\mathbf{E} \cdot \mathbf{B})^2$; solving its quadratic in z^2 gives (2)–(3). Cayley–Hamilton then verifies every identity in (4). $\square$

2 Exact velocity and position

Write all quotients below by analytic continuation at $a = 0$ or $b = 0$.

Theorem 2 (closed flow). *For $D > 0$,*

$$e^{\tau A} = P_h \left(\cosh(a\tau) I + \frac{\sinh(a\tau)}{a} A \right) + P_r \left(\cos(b\tau) I + \frac{\sin(b\tau)}{b} A \right), \quad (5)$$

$$\begin{aligned} \Phi(\tau) = & P_h \left(\frac{\sinh(a\tau)}{a} I + \frac{\cosh(a\tau) - 1}{a^2} A \right) \\ & + P_r \left(\frac{\sin(b\tau)}{b} I + \frac{1 - \cos(b\tau)}{b^2} A \right). \end{aligned} \quad (6)$$

The solution is $u(\tau) = e^{\tau A} u_0$ and $x(\tau) = x_0 + \Phi(\tau) u_0$. Moreover $\eta(u, u) = \eta(u_0, u_0)$ and $\det e^{\tau A} = 1$.

Proof. Functional calculus on (4) gives (5), and direct integration gives (6). Since $A^T \eta + \eta A = 0$, differentiation of $e^{\tau A^T} \eta e^{\tau A}$ gives zero. Finally $\det e^{\tau A} = e^{\tau \text{tr} A} = 1$. $\square$

3 Period, nonclosure, and boundary atlas

On P_h the flow has real exponents $\pm a$; on P_r it rotates with frequency b . Hence a nonconstant $u(\tau)$ is periodic exactly when

$$b > 0, \quad P_r u_0 \neq 0, \quad \text{and} \quad (a = 0 \text{ or } P_h u_0 = 0), \quad (7)$$

and its least period is $2\pi/b$. Notice the magnetic face: when $a = 0 < b$, $P_h u_0$ is constant and does not obstruct periodicity. When $a > 0$, a nonzero hyperbolic component does. If $b = 0$, no nonconstant velocity is periodic.

For a future-timelike physical initial velocity, the flow in (5) stays in the identity component of the Lorentz group. Thus $u^0(\tau) > 0$, so $x^0(\tau)$ is strictly increasing. No physical worldline closes, even when its velocity is periodic. This is a proper-time statement; it is not a claim of periodicity in coordinate time.

If $a, b > 0$, (5)–(6) split motion into hyperbolic and rotational planes. For $a > 0 = b$ (electric-like), the rotation plane becomes a kernel and (6) has a secular τP_r term. For $a = 0 < b$ (magnetic-like), the boost plane becomes a kernel and produces τP_h , alongside the cyclotron rotation. At $A = 0$, $u = u_0$ and $x = x_0 + \tau u_0$.

References

- [1] S. A. Chin, *Relativistic Motion in a Constant Electromagnetic Field*, arXiv:0809.0859 [math-ph] (2008), [doi:10.1063/1.3064796](https://doi.org/10.1063/1.3064796).